# BİL 455/533 – Parallel Processing

## Exercise #3

1. **Histogram computation with CUDA.** You can use the OpenCV's C++ API to read-in images. and access pixel values. You can use the provided image lena_grayscale_hq.jpg.
   i. Write a multi-threaded C++ code that computes the histograms of 8-bit grayscale images. Optimize this code, pick the best number of threads, etc.
   ii. Write a CUDA/C++ code that computes the histograms of 8-bit grayscale images.
   iii. Compare the execution times of the codes in parts (i) and (ii).

2. **Integral array computation with CUDA.** You can assume 8-bit int data type.
   i. Write a multi-threaded C++ code that computes the integral arrays of int arrays. Optimize this code, pick the best number of threads, etc.
   ii. Write a CUDA/C++ code that computes the integral arrays of int arrays. Use the standard C++ threads.
   iii. Compare the execution times of the codes in parts (i) and (ii) for varying array sizes.

3. **Integral image computation with CUDA.** You can use the OpenCV's C++ API to read-in images. and access pixel values. You can use the provided image lena_grayscale_hq.jpg.
   i. Write a multi-threaded C++ code that computes the integral images of 8-bit grayscale images. Optimize this code, pick the best number of threads, etc.
   ii. Write a CUDA/C++ code that computes the integral images of 8-bit grayscale images.
   iii. Compare the execution times of the codes in parts (i) and (ii) for varying image sizes. You can use OpenCV's C++ API to resize your image to obtain "more" pixels…